

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396853
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Colli; Andrea et al.

Systems and methods for delivery of chordae replacement system

Abstract

A system and method of repairing a native chordae of a patient using an artificial chordae. The artificial chordae is inserted into the patient in a non-deployed configuration using a delivery system and is delivered to a desired position within the patient. An operator causes the artificial chordae to transition from the non-deployed configuration to a deployed configuration. The artificial chordae is anchored to a myocardium of the patient. The artificial chordae is attached to a leaflet of the native chordae at an attachment location and tuned to a desired tension.

Inventors:	Colli; Andrea (Padua, IT), Yellin; Nadav (Aven Yehuda, IL), Shaolian; Samuel (Newport Beach, CA), Porath; Shuki (Haifa, IL)
Applicant:	VALCARE MEDICAL, INC. (Wilmington, DE)
Family ID:	1000008780522
Assignee:	Valcare Medical, Inc. (Wilmington, DE)
Appl. No.:	17/595919
Filed (or PCT Filed):	June 11, 2020
PCT No.:	PCT/US2020/037296
PCT Pub. No.:	WO2020/252202
PCT Pub. Date:	December 17, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20220226117 A1	Jul. 21, 2022

Related U.S. Application Data

Publication Classification**Int. Cl.:** **A61F2/24** (20060101); **A61B17/00** (20060101)**U.S. Cl.:****CPC** **A61F2/2466** (20130101); **A61B17/00234** (20130101); **A61F2/2457** (20130101);
A61B2017/00243 (20130101); **A61F2220/0016** (20130101); **A61F2220/0041** (20130101)**Field of Classification Search****CPC:** A61F (2/2466); A61F (2/2457); A61F (2220/0016); A61F (2220/0041); A61F (2250/0007); A61B (17/00234); A61B (2017/00243); A61B (17/122); A61B (2017/00783); A61B (2017/0404); A61B (2017/0406); A61B (2017/0417); A61B (2017/0437); A61B (2017/0441); A61B (2017/0464); A61B (2017/0495); A61B (2017/0496); A61B (17/0401); A61B (2017/0435); A61B (2017/0412); A61B (2017/0445); A61B (2017/0427)

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4602911	12/1985	Ahmadi et al.	N/A	N/A
4874378	12/1988	Hillstead	N/A	N/A
4953540	12/1989	Ray et al.	N/A	N/A
5080662	12/1991	Paul	N/A	N/A
5236440	12/1992	Hlavacek	N/A	N/A
5306296	12/1993	Wright et al.	N/A	N/A
5609565	12/1996	Nakamura	N/A	N/A
5695518	12/1996	Laerum	N/A	N/A
5716370	12/1997	Williamson, IV et al.	N/A	N/A
5855614	12/1998	Stevens et al.	N/A	N/A
6113611	12/1999	Allen et al.	N/A	N/A
6231602	12/2000	Carpentier et al.	N/A	N/A
6447524	12/2001	Knodel et al.	N/A	N/A
6619291	12/2002	Hlavka et al.	N/A	N/A
6629534	12/2002	St. Goar et al.	N/A	N/A
6669687	12/2002	Saadat	N/A	N/A
6689048	12/2003	Vanden Hoek et al.	N/A	N/A
6726704	12/2003	Loshakove et al.	N/A	N/A
6776784	12/2003	Ginn	N/A	N/A
6790229	12/2003	Berrekouw	N/A	N/A
6797002	12/2003	Spence et al.	N/A	N/A
6805711	12/2003	Quijano et al.	N/A	N/A

6869444	12/2004	Gabbay	N/A	N/A
6893459	12/2004	MacOviak	N/A	N/A
7101395	12/2005	Tremulis et al.	N/A	N/A
7114953	12/2005	Wagner	N/A	N/A
7175660	12/2006	Cartledge et al.	N/A	N/A
7238191	12/2006	Bachmann	N/A	N/A
7285087	12/2006	Moaddeb et al.	N/A	N/A
7297150	12/2006	Cartledge et al.	N/A	N/A
7569072	12/2008	Berg et al.	N/A	N/A
7594887	12/2008	Moaddeb et al.	N/A	N/A
7635329	12/2008	Goldfarb et al.	N/A	N/A
7655040	12/2009	Douk et al.	N/A	N/A
7717954	12/2009	Solem et al.	N/A	N/A
7722668	12/2009	Moaddeb et al.	N/A	N/A
7758637	12/2009	Starksen et al.	N/A	N/A
7828819	12/2009	Webler et al.	N/A	N/A
7837729	12/2009	Gordon et al.	N/A	N/A
7988725	12/2010	Gross et al.	N/A	N/A
8163014	12/2011	Lane et al.	N/A	N/A
8182529	12/2011	Gordon et al.	N/A	N/A
8236049	12/2011	Rowe et al.	N/A	N/A
8287591	12/2011	Keidar et al.	N/A	N/A
8333204	12/2011	Saadat	606/232	A61B 18/1442
8518107	12/2012	Tsukashima et al.	N/A	N/A
8579968	12/2012	Shannon et al.	N/A	N/A
8690939	12/2013	Miller	623/2.11	A61F 2/2457
8821570	12/2013	DuMontelle et al.	N/A	N/A
9180008	12/2014	Yellin et al.	N/A	N/A
9402721	12/2015	Buchbinder et al.	N/A	N/A
9433503	12/2015	Tsukashima et al.	N/A	N/A
9839519	12/2016	Shaolian et al.	N/A	N/A
9877833	12/2017	Bishop et al.	N/A	N/A
10405979	12/2018	Schaffner	N/A	A61B 17/0469
10543087	12/2019	Yellin et al.	N/A	N/A
10779945	12/2019	Buchbinder et al.	N/A	N/A
11058417	12/2020	Foerster et al.	N/A	N/A
11191536	12/2020	Foerster et al.	N/A	N/A
11224422	12/2021	Foerster et al.	N/A	N/A
11298230	12/2021	Shaolian et al.	N/A	N/A
11382749	12/2021	Yellin et al.	N/A	N/A
11510835	12/2021	Yellin et al.	N/A	N/A
11534300	12/2021	Yellin et al.	N/A	N/A
11571301	12/2022	Yellin et al.	N/A	N/A
11571307	12/2022	Yellin et al.	N/A	N/A
11576779	12/2022	Yellin et al.	N/A	N/A
11617647	12/2022	Yellin	N/A	N/A
11654018	12/2022	Shaolian et al.	N/A	N/A

11793628	12/2022	Dumontelle et al.	N/A	N/A
11806009	12/2022	Foerster et al.	N/A	N/A
11806237	12/2022	Rozen et al.	N/A	N/A
11813164	12/2022	Yellin et al.	N/A	N/A
11857418	12/2023	Yellin et al.	N/A	N/A
12115069	12/2023	Shaolian et al.	N/A	N/A
12127941	12/2023	Yellin et al.	N/A	N/A
2002/0151961	12/2001	Lashinski et al.	N/A	N/A
2002/0151970	12/2001	Garrison et al.	N/A	N/A
2002/0188170	12/2001	Santamore	623/23.64	A61K 9/14
2002/0198526	12/2001	Shaolian et al.	N/A	N/A
2003/0050649	12/2002	Brock et al.	N/A	N/A
2003/0050693	12/2002	Quijano et al.	N/A	N/A
2003/0078465	12/2002	Pai et al.	N/A	N/A
2003/0078671	12/2002	Lesniak et al.	N/A	N/A
2003/0191528	12/2002	Quijano et al.	N/A	N/A
2003/0198605	12/2002	Montgomery	N/A	N/A
2003/0199974	12/2002	Lee et al.	N/A	N/A
2004/0044364	12/2003	DeVries et al.	N/A	N/A
2004/0068276	12/2003	Golden et al.	N/A	N/A
2004/0073237	12/2003	Leinsing	N/A	N/A
2004/0122514	12/2003	Fogarty et al.	N/A	N/A
2004/0138744	12/2003	Lashinski et al.	N/A	N/A
2004/0148021	12/2003	Cartledge et al.	N/A	N/A
2004/0193191	12/2003	Starksen et al.	N/A	N/A
2004/0243230	12/2003	Navia et al.	N/A	N/A
2004/0249391	12/2003	Cummins	N/A	N/A
2004/0260393	12/2003	Rahdert et al.	N/A	N/A
2004/0260394	12/2003	Douk et al.	N/A	N/A
2005/0004668	12/2004	Aklog et al.	N/A	N/A
2005/0020696	12/2004	Montgomery et al.	N/A	N/A
2005/0033325	12/2004	May et al.	N/A	N/A
2005/0065550	12/2004	Starksen et al.	N/A	N/A
2005/0090846	12/2004	Pedersen et al.	N/A	N/A
2005/0096740	12/2004	Langberg et al.	N/A	N/A
2005/0113910	12/2004	Paniagua et al.	N/A	N/A
2005/0137692	12/2004	Haug et al.	N/A	N/A
2005/0137695	12/2004	Salahieh et al.	N/A	N/A
2005/0203549	12/2004	Realyvasquez	N/A	N/A
2005/0222678	12/2004	Lashinski et al.	N/A	N/A
2005/0240200	12/2004	Bergheim	N/A	N/A
2005/0250161	12/2004	Suciu-Foca et al.	N/A	N/A
2005/0267572	12/2004	Schoon et al.	N/A	N/A
2005/0283190	12/2004	Huitema et al.	N/A	N/A
2005/0288778	12/2004	Shaoulian et al.	N/A	N/A
2005/0288781	12/2004	Moaddeb et al.	N/A	N/A
2006/0009737	12/2005	Whiting et al.	N/A	N/A
2006/0020327	12/2005	Lashinski et al.	N/A	N/A
2006/0052867	12/2005	Revuelta et al.	N/A	N/A
2006/0122633	12/2005	To et al.	N/A	N/A

2006/0129025	12/2005	Levine et al.	N/A	N/A
2006/0155165	12/2005	Vanden Hoek et al.	N/A	N/A
2006/0161169	12/2005	Nieminen et al.	N/A	N/A
2006/0184240	12/2005	Jimenez et al.	N/A	N/A
2006/0184242	12/2005	Lichtenstein	N/A	N/A
2006/0195134	12/2005	Crittenden	N/A	N/A
2006/0195183	12/2005	Navia et al.	N/A	N/A
2006/0241748	12/2005	Lee et al.	N/A	N/A
2006/0282161	12/2005	Huynh et al.	N/A	N/A
2007/0016287	12/2006	Cartledge et al.	N/A	N/A
2007/0027533	12/2006	Douk	N/A	N/A
2007/0038296	12/2006	Navia	N/A	N/A
2007/0051377	12/2006	Douk et al.	N/A	N/A
2007/0067027	12/2006	Moaddeb et al.	N/A	N/A
2007/0073098	12/2006	Lenker et al.	N/A	N/A
2007/0080188	12/2006	Spence et al.	N/A	N/A
2007/0093854	12/2006	Kayan	N/A	N/A
2007/0118215	12/2006	Moaddeb	N/A	N/A
2007/0128132	12/2006	Piergallini et al.	N/A	N/A
2007/0135913	12/2006	Moaddeb et al.	N/A	N/A
2007/0142907	12/2006	Moaddeb et al.	N/A	N/A
2007/0213812	12/2006	Webler et al.	N/A	N/A
2007/0233239	12/2006	Navia et al.	N/A	N/A
2007/0239272	12/2006	Navia et al.	N/A	N/A
2007/0244553	12/2006	Rafiee et al.	N/A	N/A
2007/0244554	12/2006	Rafiee et al.	N/A	N/A
2007/0244555	12/2006	Rafiee et al.	N/A	N/A
2007/0244556	12/2006	Rafiee et al.	N/A	N/A
2007/0250161	12/2006	Dolan	N/A	N/A
2007/0293942	12/2006	Mirzaee	N/A	N/A
2008/0039934	12/2007	Styrc	N/A	N/A
2008/0051807	12/2007	St. Goar et al.	N/A	N/A
2008/0065011	12/2007	Marchand et al.	N/A	N/A
2008/0177380	12/2007	Starksen et al.	N/A	N/A
2008/0177381	12/2007	Navia et al.	N/A	N/A
2008/0200980	12/2007	Robin et al.	N/A	N/A
2008/0208327	12/2007	Rowe	N/A	N/A
2008/0215145	12/2007	Moaddeb et al.	N/A	N/A
2008/0243220	12/2007	Barker	N/A	N/A
2008/0262513	12/2007	Stahler et al.	N/A	N/A
2008/0262609	12/2007	Gross et al.	N/A	N/A
2008/0288060	12/2007	Kaye et al.	N/A	N/A
2008/0306586	12/2007	Cartledge et al.	N/A	N/A
2009/0076599	12/2008	Bergin	N/A	N/A
2009/0088838	12/2008	Shaolian et al.	N/A	N/A
2009/0118747	12/2008	Bettuchi et al.	N/A	N/A
2009/0125098	12/2008	Chuter	N/A	N/A
2009/0149872	12/2008	Gross et al.	N/A	N/A
2009/0216322	12/2008	Le et al.	N/A	N/A

2009/0222083	12/2008	Nguyen et al.	N/A	N/A
2009/0238778	12/2008	Mordas et al.	N/A	N/A
2009/0299470	12/2008	Rao et al.	N/A	N/A
2010/0010616	12/2009	Drews et al.	N/A	N/A
2010/0030014	12/2009	Ferrazzi	N/A	N/A
2010/0063586	12/2009	Hasenkam et al.	N/A	N/A
2010/0076549	12/2009	Keidar et al.	N/A	N/A
2010/0121433	12/2009	Bolling et al.	N/A	N/A
2010/0161047	12/2009	Cabiri	N/A	N/A
2010/0185274	12/2009	Moaddeb et al.	N/A	N/A
2010/0191327	12/2009	Lane et al.	N/A	N/A
2010/0204713	12/2009	Ruiz Morales	N/A	N/A
2010/0211166	12/2009	Miller et al.	N/A	N/A
2010/0249920	12/2009	Bolling et al.	N/A	N/A
2010/0262232	12/2009	Annest	N/A	N/A
2010/0266989	12/2009	Piergallini et al.	N/A	N/A
2010/0280604	12/2009	Zipory et al.	N/A	N/A
2010/0280605	12/2009	Hammer et al.	N/A	N/A
2010/0286767	12/2009	Zipory et al.	N/A	N/A
2011/0004299	12/2010	Navia et al.	N/A	N/A
2011/0011917	12/2010	Loulmet	N/A	N/A
2011/0022168	12/2010	Cartledge	N/A	N/A
2011/0027753	12/2010	Maurat et al.	N/A	N/A
2011/0034953	12/2010	Milo	N/A	N/A
2011/0034999	12/2010	Carpentier et al.	N/A	N/A
2011/0060407	12/2010	Ketai	623/2.37	A61B 17/0644
2011/0066231	12/2010	Cartledge et al.	N/A	N/A
2011/0093062	12/2010	Cartledge et al.	N/A	N/A
2011/0106245	12/2010	Miller et al.	N/A	N/A
2011/0106247	12/2010	Miller et al.	N/A	N/A
2011/0137397	12/2010	Chau et al.	N/A	N/A
2011/0166649	12/2010	Gross et al.	N/A	N/A
2011/0190879	12/2010	Bobo et al.	N/A	N/A
2011/0208295	12/2010	Cartledge et al.	N/A	N/A
2011/0208298	12/2010	Tuval et al.	N/A	N/A
2011/0224785	12/2010	Hacohen	N/A	N/A
2011/0257728	12/2010	Kuehn	N/A	N/A
2011/0282361	12/2010	Miller et al.	N/A	N/A
2011/0301698	12/2010	Miller et al.	N/A	N/A
2011/0301699	12/2010	Saadat	N/A	N/A
2011/0319989	12/2010	Lane et al.	N/A	N/A
2012/0022557	12/2011	Cabiri et al.	N/A	N/A
2012/0022639	12/2011	Hacohen	623/2.11	A61F 2/2436
2012/0022644	12/2011	Reich et al.	N/A	N/A
2012/0059458	12/2011	Buchbinder et al.	N/A	N/A
2012/0083806	12/2011	Goertzen	606/151	A61B 17/0401
2012/0083880	12/2011	Rankin et al.	N/A	N/A

2012/0095455	12/2011	Rodmond et al.	N/A	N/A
2012/0123531	12/2011	Tsukashima	623/2.37	A61F 2/2448
2012/0136436	12/2011	Cabiri et al.	N/A	N/A
2012/0136463	12/2011	Muniz	N/A	N/A
2012/0165930	12/2011	Gifford, III et al.	N/A	N/A
2012/0245604	12/2011	Tegzes	N/A	N/A
2012/0310330	12/2011	Buchbinder et al.	N/A	N/A
2013/0066342	12/2012	Dell et al.	N/A	N/A
2013/0087598	12/2012	Surti	N/A	N/A
2013/0116780	12/2012	Miller et al.	N/A	N/A
2013/0166022	12/2012	Conklin	N/A	N/A
2013/0204361	12/2012	Adams et al.	N/A	N/A
2013/0226289	12/2012	Shaolian	623/2.37	A61F 2/2448
2013/0226290	12/2012	Yellin et al.	N/A	N/A
2013/0282114	12/2012	Schweich, Jr. et al.	N/A	N/A
2013/0289718	12/2012	Tsukashima et al.	N/A	N/A
2013/0289720	12/2012	Dobrilovic	N/A	N/A
2013/0304197	12/2012	Buchbinder et al.	N/A	N/A
2014/0005778	12/2013	Buchbinder et al.	N/A	N/A
2014/0025163	12/2013	Padala et al.	N/A	N/A
2014/0046433	12/2013	Kovalsky	N/A	N/A
2014/0058505	12/2013	Bielefeld	N/A	N/A
2014/0114407	12/2013	Rajamannan	N/A	N/A
2014/0188130	12/2013	Sanchez et al.	N/A	N/A
2014/0214157	12/2013	Börtlein et al.	N/A	N/A
2014/0277411	12/2013	Börtlein et al.	N/A	N/A
2014/0364939	12/2013	Deshmukh et al.	N/A	N/A
2015/0073420	12/2014	Bookwalter et al.	N/A	N/A
2015/0112432	12/2014	Reich et al.	N/A	N/A
2015/0127093	12/2014	Hosmer et al.	N/A	N/A
2015/0173897	12/2014	Raanani et al.	N/A	N/A
2015/0173987	12/2014	Albinmousa et al.	N/A	N/A
2015/0351903	12/2014	Morriss et al.	N/A	N/A
2015/0366666	12/2014	Khairkhahan et al.	N/A	N/A
2016/0022419	12/2015	Yellin et al.	N/A	N/A
2016/0038286	12/2015	Yellin et al.	N/A	N/A
2016/0089235	12/2015	Yellin	N/A	N/A
2016/0100897	12/2015	Avalos et al.	N/A	N/A
2016/0106420	12/2015	Foerster et al.	N/A	N/A
2016/0120642	12/2015	Shaolian et al.	N/A	N/A
2016/0120645	12/2015	Alon	N/A	N/A
2016/0220371	12/2015	Keane	N/A	N/A
2016/0270911	12/2015	Ganesan et al.	N/A	N/A
2016/0331534	12/2015	Buchbinder et al.	N/A	N/A
2017/0042670	12/2016	Shaolian et al.	N/A	N/A
2017/0231763	12/2016	Yellin	N/A	N/A
2017/0258590	12/2016	Khairkhahan	N/A	N/A

2017/0296340	12/2016	Gross et al.	N/A	N/A
2017/0325948	12/2016	Wallace et al.	N/A	N/A
2018/0028387	12/2017	Yellin et al.	N/A	N/A
2018/0042723	12/2017	Yellin et al.	N/A	N/A
2018/0098849	12/2017	Yellin et al.	N/A	N/A
2018/0161160	12/2017	Shaolian et al.	N/A	N/A
2018/0161161	12/2017	Yellin et al.	N/A	N/A
2018/0228610	12/2017	Lashinski et al.	N/A	N/A
2018/0235758	12/2017	Biadillah et al.	N/A	N/A
2018/0325670	12/2017	De Canniere	N/A	N/A
2019/0053905	12/2018	Alon	N/A	N/A
2019/0083091	12/2018	Foerster et al.	N/A	N/A
2019/0083092	12/2018	Foerster et al.	N/A	N/A
2019/0083239	12/2018	Shaolian et al.	N/A	N/A
2019/0083240	12/2018	Shaolian et al.	N/A	N/A
2019/0091022	12/2018	Yellin et al.	N/A	N/A
2019/0133765	12/2018	Yellin et al.	N/A	N/A
2020/0069426	12/2019	Conklin	N/A	A61F 2/2487
2020/0163763	12/2019	Zipory et al.	N/A	N/A
2020/0170799	12/2019	Yellin et al.	N/A	N/A
2020/0330228	12/2019	Anderson et al.	N/A	N/A
2021/0085463	12/2020	Yellin et al.	N/A	N/A
2021/0161662	12/2020	Albes	N/A	N/A
2021/0353417	12/2020	Yellin et al.	N/A	N/A
2022/0226116	12/2021	Colli et al.	N/A	N/A
2022/0226771	12/2021	Lipscomb	N/A	N/A
2023/0040083	12/2022	Gifford, III et al.	N/A	N/A
2023/0045532	12/2022	Galler et al.	N/A	N/A
2023/0372086	12/2022	Galler et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2114422	12/1991	CN	N/A
2633218	12/2003	CN	N/A
101411632	12/2008	CN	N/A
101460113	12/2008	CN	N/A
101553190	12/2008	CN	N/A
102014797	12/2010	CN	N/A
102088930	12/2010	CN	N/A
202859386	12/2012	CN	N/A
103179920	12/2012	CN	N/A
103237523	12/2012	CN	N/A
103735337	12/2013	CN	N/A
203954080	12/2013	CN	N/A
108618871	12/2017	CN	N/A
113855324	12/2020	CN	N/A
102014102653	12/2014	DE	N/A
1752115	12/2006	EP	N/A

2471464	12/2011	EP	N/A
2600799	12/2012	EP	N/A
2928538	12/2014	EP	N/A
2967700	12/2015	EP	N/A
2600799	12/2016	EP	N/A
3213715	12/2016	EP	N/A
2928538	12/2017	EP	N/A
2845889	12/2003	FR	N/A
1496804	12/1977	GB	N/A
2366319	12/2001	GB	N/A
20040095482	12/2003	KR	N/A
125062	12/2012	RU	N/A
WO-8000673	12/1979	WO	N/A
WO-9009153	12/1989	WO	N/A
WO-9728745	12/1996	WO	N/A
WO-03017874	12/2002	WO	N/A
WO-03047467	12/2002	WO	N/A
WO-2005046488	12/2004	WO	N/A
WO-2007035882	12/2006	WO	N/A
WO-2008097999	12/2007	WO	N/A
WO-2009052427	12/2008	WO	N/A
WO-2009120764	12/2008	WO	N/A
WO-2010004546	12/2009	WO	N/A
WO-2010085659	12/2009	WO	N/A
WO-2011011443	12/2010	WO	N/A
WO-2011097355	12/2010	WO	N/A
2011154942	12/2010	WO	N/A
WO-2012004679	12/2011	WO	N/A
WO-2012019052	12/2011	WO	N/A
WO-2012038550	12/2011	WO	N/A
2012040865	12/2011	WO	N/A
WO-2012063228	12/2011	WO	N/A
WO-2012095159	12/2011	WO	N/A
WO-2012106354	12/2011	WO	N/A
WO-2012167095	12/2011	WO	N/A
WO-2013095816	12/2012	WO	N/A
WO-2013128436	12/2012	WO	N/A
WO-2013130641	12/2012	WO	N/A
WO-2013175468	12/2012	WO	N/A
WO-2014089424	12/2013	WO	N/A
WO-2014145399	12/2013	WO	N/A
WO-2014178869	12/2013	WO	N/A
WO-2014189509	12/2013	WO	N/A
WO-2014190329	12/2013	WO	N/A
WO-2014210600	12/2013	WO	N/A
WO-2015052629	12/2014	WO	N/A
WO-2015132668	12/2014	WO	N/A
WO-2016025894	12/2015	WO	N/A
WO-2016040526	12/2015	WO	N/A
WO-2018035118	12/2017	WO	N/A

WO-2018071540	12/2017	WO	N/A
WO-2018170424	12/2017	WO	N/A
WO-2020117842	12/2019	WO	N/A
WO-2020252200	12/2019	WO	N/A

OTHER PUBLICATIONS

PCT/US2020/037296, International Search Report and Written Opinion, 7 pages, Sep. 10, 2020.
cited by applicant

Extended European Search Report for European Application No. 11815347.7, mailed Mar. 14, 2016, 10 Pages. cited by applicant

Extended European Search Report for European Application No. 12793292.9, mailed Dec. 1, 2014, 6 Pages. cited by applicant

Extended European Search Report for European Application No. 13755441.6, mailed Mar. 1, 2016, 12 Pages. cited by applicant

Extended European Search Report for European Application No. 13860442.6, mailed Aug. 11, 2016, 7 pages. cited by applicant

Extended European Search Report for European Application No. 13885021.9, mailed Jan. 3, 2017, 8 Pages. cited by applicant

Extended European Search Report for European Application No. 14762806.9, mailed Jul. 29, 2016, 7 Pages. cited by applicant

Extended European Search Report for European Application No. 14801009.3, mailed Dec. 5, 2016, 8 Pages. cited by applicant

Extended European Search Report for European Application No. 14817662.1, mailed Jan. 23, 2017, 7 Pages. cited by applicant

Extended European Search Report for European Application No. 17155803.4, mailed Aug. 9, 2017, 10 Pages. cited by applicant

Extended European Search Report for European Application No. 17835256.3, mailed Feb. 12, 2020, 9 Pages. cited by applicant

Extended European Search Report for European Application No. 17841988.3, mailed Dec. 16, 2019, 8 Pages. cited by applicant

Extended European Search Report for European Application No. 17860901.2, mailed Jun. 5, 2020, 06 Pages. cited by applicant

Extended European Search Report for European Application No. 18768197.8, mailed Oct. 19, 2020, 7 Pages. cited by applicant

Extended European Search Report for European Application No. 19151726.7, mailed Jul. 22, 2019, 9 Pages. cited by applicant

Extended European Search Report for European Application No. 19170261.2, mailed Aug. 5, 2019, 9 pages. cited by applicant

Extended European Search Report for European Application No. 19893113.1, mailed Nov. 17, 2022, 7 Pages. cited by applicant

Extended European Search Report for European Application No. 20206790.6, mailed Dec. 7, 2020, 9 Pages. cited by applicant

Extended European Search Report for European Application No. 20209605.3, mailed Mar. 9, 2021, 7 pages. cited by applicant

Extended European Search Report for European Application No. 20823198.5, mailed May 15, 2023, 15 Pages. cited by applicant

Extended European Search Report for European Application No. 20823682.8, mailed Apr. 14, 2023, 10 Pages. cited by applicant

Extended European Search Report for European Application No. 20841346.8, mailed Jul. 21, 2023, 9 Pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/IL2022/050868, mailed Nov. 17, 2022, 19 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/IL2023/050527, mailed Aug. 8, 2023, 14 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2011/046659, mailed Jun. 4, 2012, 13 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2012/040481, mailed Dec. 6, 2012, 12 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2013/028065, mailed Jun. 27, 2013, 11 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2013/042275, mailed Feb. 20, 2014, 18 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2013/058102, mailed Apr. 21, 2014, 10 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2013/073552, mailed Mar. 6, 2014, 10 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2014/030163, mailed Aug. 27, 2014, 10 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2014/039454, mailed Oct. 22, 2014, 11 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2014/044920, mailed Dec. 24, 2014, 15 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2017/044129, mailed Sep. 27, 2017, 9 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2017/046933, mailed Dec. 21, 2017, 11 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2017/056138, mailed Jan. 8, 2018, 6 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2018/022910, mailed May 23, 2018, 7 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2019/064289, mailed Feb. 5, 2020, 8 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2020/037294, mailed Aug. 28, 2020, 8 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2020/042201, mailed Oct. 9, 2020, 7 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2021/071467, mailed Jan. 14, 2022, 12 Pages. cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2021/071468, mailed Jan. 19, 2022, 8 Pages. cited by applicant
Partial Supplementary European Search Report for European Application No. 11815347.7, mailed Nov. 16, 2015, 06 Pages. cited by applicant
Partial Supplementary European Search Report for European Application No. 13755441.6, mailed Nov. 3, 2015, 7 Pages. cited by applicant
International Search Report & Written Opinion dated Jul. 24, 2024 in Int'l PCT Patent Appl. Serial No. PCT/US2024/019797. cited by applicant

Background/Summary

PRIORITY (1) The present application is a national stage filing under 35 U.S.C. § 371 of International Patent Application No. PCT/US2020/037296, filed Jun. 11, 2020, which claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 62/859,968, filed on Jun. 11, 2019, titled “SYSTEMS AND METHODS FOR DELIVERY OF CHORDEA REPLACEMENT SYSTEM,” each of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

(1) The present invention relates to implantable devices. The invention is particularly useful in devices implantable by catheter for the treatment of mitral or tricuspid regurgitation. The cause of the regurgitation can be functional, degenerative, or any other reason. The invention could be used for valvular lesions as well.

(2) Mitral regurgitation is a valvular dysfunction that causes blood volume to flow during systole (i.e., left ventricular contraction) from the left ventricle to the left atrium, in contrast to a healthy heart where this direction of flow is blocked by the mitral valve. The reversed flow during systole causes a rise in pressure in the left atrium. Maintaining a normal cardiac output results in increased left ventricle pressure.

(3) Treating patients with mitral or tricuspid regurgitation could require valve repair or replacement in order to reduce or eliminate the regurgitation. For many years, the commonly accepted treatment was surgical repair or replacement of the native valve during open heart surgery. Valve repair is a procedure that may require complementary treatments such as utilizing annuloplasty rings with or without leaflet or chordae repair. In recent years, a trans vascular technique has been developed for introducing these devices in general and implanting an artificial chorda through a catheter in a manner that is less invasive than open heart surgery.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects, features, benefits and advantages of the embodiments described herein will be apparent with regard to the following description, appended claims, and accompanying drawings where:

(2) FIG. 1 depicts a short axis view of the heart in accordance with an embodiment.

(3) FIG. 2 depicts mitral valve leaflets of the heart in accordance with an embodiment.

(4) FIG. 3 depicts a long axis anterior to posterior view of the heart in accordance with an embodiment.

(5) FIG. 4 depicts a long axis commissure to commissure view of the heart in accordance with an embodiment.

(6) FIG. 5 depicts an illustrative transatrial approach in a commissure to commissure view of the heart in accordance with an embodiment.

(7) FIG. 6 depicts an illustrative transatrial approach in an anterior to posterior view of the heart in accordance with an embodiment.

(8) FIG. 7 depicts an illustrative transapical approach in a commissure to commissure view of the heart in accordance with an embodiment.

- (9) FIG. 8 depicts an illustrative transfemoral approach in an anterior to posterior view of the heart in accordance with an embodiment.
- (10) FIG. 9 depicts an illustrative posterior leaflet with a plurality of artificial chordae being implanted in accordance with an embodiment.
- (11) FIG. 10 depicts a side view of a posterior leaflet with an artificial chordae being implanted in accordance with an embodiment.
- (12) FIG. 11 depicts an illustrative artificial chordae with a leaflet capture mechanism in accordance with an embodiment.
- (13) FIG. 12 depicts an illustrative open leaflet capture configuration on an artificial chordae in accordance with an embodiment.
- (14) FIG. 13 depicts an illustrative closed leaflet capture configuration on an artificial chordae in accordance with an embodiment.
- (15) FIG. 14 depicts an illustrative pulley-based attachment and tuning mechanism in accordance with an embodiment.
- (16) FIG. 15 depicts an illustrative knot-based attachment and tuning mechanism in accordance with an embodiment.
- (17) FIG. 16 depicts an illustrative knot-based attachment and tuning mechanism in accordance with an embodiment.
- (18) FIG. 17 depicts another illustrative anchor-based attachment and tuning mechanism in accordance with an embodiment.
- (19) FIG. 18 depicts an illustrative anchor subassembly for an artificial chordae in accordance with an embodiment.
- (20) FIG. 19 depicts an illustrative inner member of an anchor subassembly in a deployed configuration in accordance with an embodiment.
- (21) FIG. 20 depicts an illustrative inner member of an anchor subassembly in a non-deployed configuration in accordance with an embodiment.
- (22) FIG. 21 depicts an illustrative outer member of an anchor subassembly in accordance with an embodiment.
- (23) FIG. 22 depicts an illustrative anchor subassembly in a non-deployed configuration in accordance with an embodiment.
- (24) FIG. 23 depicts an illustrative anchor subassembly in a deployed configuration in accordance with an embodiment.

DETAILED DESCRIPTION

- (25) This disclosure is not limited to the particular systems, devices, and methods described, as these may vary. The terminology used in the description is for the purpose of describing the particular versions or embodiments only, and is not intended to limit the scope.
- (26) As used in this document, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art. Nothing in this disclosure is to be construed as an admission that the embodiments described in this disclosure are not entitled to antedate such disclosure by virtue of prior invention. As used in this document, the term “comprising” means “including, but not limited to.”
- (27) When the term “valvular apparatus” is used in different variations, it may refer to the mitral valve apparatus and/or to the tricuspid valve apparatus and include the leaflets, the chordae, and/or the papillary muscles.
- (28) In an embodiment, an artificial chordae device is delivered to the target site, the mitral or tricuspid valve apparatus, through a catheter while the artificial chordae and their anchoring accessories are housed within a low diameter shaft which is correctly positioned and anchored to the functional configuration geometry and location. The catheter may be advanced to the target site through the vascular system. In an embodiment, the catheter may be advanced from the femoral

vein or artery. In alternate embodiments, the catheter may be advanced from any blood vessel that allows access to the target site. In some embodiments the catheter may be advanced transapically, where a catheter is advanced through a small incision made in the chest wall and then through the apex. In some embodiments, the catheter may be advanced transatrially, where a catheter is advanced through a small incision made in the chest wall and then through the left or right atrium. (29) The artificial chordae device may include one or more of a variety of attachment methods for attachment to the native leaflet and myocardium. The artificial chordae device may include one or more of a variety of locking mechanisms between the leaflet capture subassembly and the native leaflet. In some embodiments, the artificial chordae device may include barbs to prevent rocking and device movement in relation to the leaflet upon insertion. The artificial chordae device may include one or more of a variety of tuning mechanisms. In addition, the artificial chordae device may include one or more of a variety of features that lock the tuning mechanism. Examples of the above-identified variations will be described further herein, although the described variations are illustrative only and are not meant to be limiting.

(30) The illustrated example embodiments will be best understood by reference to the figures. The following description is intended only by way of example and illustrates certain example embodiments.

(31) FIG. 1 depicts a short axis view of the heart displaying the four valves: the mitral valve **121**, the tricuspid valve **109**, the pulmonary valve **110**, and the aortic valve **107**. As shown, the mitral valve **121** includes the posterior leaflet **104** and the anterior leaflet **105**, which will be discussed further herein in relation to various embodiments.

(32) FIG. 2 depicts a perspective of the mitral valve commonly known as the “surgical view” which identifies individual segments of the native leaflets of the mitral valve **221**. As depicted in FIG. 2, individual segments (A1, A2, A3) of the anterior leaflet **205** and individual segments (P1, P2 and P3) of the posterior leaflet **204** are identified. The lateral commissure **219** and medial commissure **220** are also noted for reference.

(33) FIG. 3 depicts a left ventricular outflow tract view, anterior to posterior, of the heart which shows how the anterior **305** and posterior **304** leaflets are attached to the papillary muscles **303** by the chordae **306**. FIG. 3 further depicts the left ventricle **302** and the right ventricle **301** separated by the intraventricular septum **312**, the apex **316**, the left atrium **308**, and the aortic valve **307**. Within the left ventricle **302**, the posterior wall **311**, mitral groove **314**, and aortic-mitral continuity **315** are specifically identified.

(34) FIG. 4 depicts a perspective of the left side of the heart in commissure to commissure view showing the left atrium **408**. Similar to FIG. 3, FIG. 4 depicts the attachment of the posterior leaflet **404** to the myocardium **403** via the chordae **406** near the apex of the heart **416**.

(35) FIG. 5 depicts an illustrative transatrial approach of a chordae repair device to the mitral valve apparatus in a commissure to commissure view of the heart in accordance with an embodiment. When using the transatrial approach, a catheter **545** is introduced into the left atrium **508** directly. In addition, the transatrial approach may pass by or cross-over the mitral valve and allow implantation of a chordae repair device between the posterior leaflet **504** and/or the papillary muscles **503** to repair the chordae **506**.

(36) FIG. 6 depicts an illustrative transatrial approach of a chordae repair device to the mitral valve apparatus in an anterior to posterior view of the heart in accordance with an embodiment. The catheter **645** is introduced into the left atrium **608** directly at the native valve comprising the anterior **605** and posterior **604** leaflets. Using this approach, implantation of aa chordae repair device may be made to the leaflets **604**, **605** and the papillary muscles **603** in repair of the chordae **606** within the left ventricle **602**. FIG. 6 further depicts the right ventricle **601** and aortic valve **607** for reference.

(37) FIG. 7 depicts an illustrative transapical approach of a chordae repair device to the mitral valve apparatus in an anterior to posterior view of the heart in accordance with an embodiment.

Using the transapical approach, the catheter **745** is introduced between the left atrium **708** and the ventricle **702**. The transapical approach is a direct approach where the catheter **745** is introduced into the left ventricle across the apex **716** to allow implantation of the chordae repair device to the valvular apparatus to anchor between the leaflet **704** (e.g., posterior leaflet, interior leaflet, etc.) and the papillary muscles **703** to repair the chordae **706**.

(38) FIG. **8** depicts an illustrative transfemoral approach of the chordae repair device to the mitral valve apparatus in an anterior to posterior view of the heart in accordance with an embodiment. The transfemoral approach is a percutaneous approach where the catheter **845** is introduced into the femoral vein and advanced along the inferior vena cava to the right atrium, across the septum **812**, to allow implantation of a chordae repair device to the valvular apparatus including the anterior **805** and posterior **804** leaflets, papillary muscles **803**, and chordae **806**. The aortic valve **807** is also shown for reference.

(39) FIG. **9** depicts an illustrative posterior leaflet with a plurality of artificial chordae being implanted in accordance with an embodiment. As shown in FIG. **9**, an artificial chordae device **922** is implanted at a plurality of locations on the posterior leaflet **904** of the left ventricle **902**. The artificial chordae device **922** may be used to repair chordae **906** that failed due to, for example and without limitation, leaflet prolapse, flail, or other pathology. In some embodiments, the artificial chordae device **922** comprises a plurality of artificial chordae **944**, an anchor subassembly **923** configured to attach to the myocardium, and a leaflet capture subassembly **935**, with a tuning mechanism **942** and stress relief pad **938**, configured to attach to the native leaflet. The implantation of one or more artificial chordae devices **922** may be performed serially or in parallel. Each artificial chordae device **922** may be tuned separately using the corresponding tuning mechanism **942**. The one or more artificial chordae devices **922** may be visualized using, for example and without limitation, endoscopy, fluoroscopy, a CT scan, an MRI scan, and/or vital signals to assist in optimizing the result.

(40) FIG. **10** depicts a side view of a posterior leaflet with an artificial chordae device being implanted in accordance with an embodiment. As shown in FIG. **10**, one or more leaflet capture assemblies **1035** may be attached to the posterior leaflet **1004** in the left ventricle **1002** from a side view. The one or more leaflet capture assemblies **1035** may include a mechanism to ensure connectivity and proper tension. In some embodiments, the connectivity and tension may be tunable (e.g., separately or in combination) during and/or after deployment and attachment. An anchor assembly **1023** may be used to attach the artificial chordae **1044** to the myocardium. The anterior leaflet **1005** and aortic valve **1007** are shown for reference.

(41) FIG. **11** depicts an illustrative artificial chordae with a leaflet capture mechanism in accordance with an embodiment. An artificial chordae device comprises an artificial chordae **1144**, a leaflet capture subassembly **1135**, and an anchor subassembly **1123**.

(42) FIGS. **12** and **13** depict the leaflet capture subassembly **1235** of the artificial chordae, which is composed of an upper jaw **1239**, a lower jaw **1240**, and a lock feature **1236** to assure proper attachment to the native leaflet. The jaws **1239**, **1240** capture the leaflet and have a depth of 2-8 mm, shown as dimension A **1241**. In an embodiment, the geometry and the attachment depth of the jaws **1239**, **1240** may be designed to prevent damage and unintentional detachment from the native leaflet. The artificial chordae **1244** may be attached to the jaws **1239**, **1240** at the tuning mechanism **1242**. The artificial chordae **1244** may be attached to a muscle using an anchor subassembly **1223**. FIG. **12** depicts the artificial chordae **1244** with the leaflet capture subassembly **1235** in an open configuration.

(43) FIG. **13** depicts the artificial chordae **1344** with the leaflet capture subassembly **1335** in a closed configuration. As shown in FIG. **13**, the locking mechanism **1336** is engaged. As shown, dimension A **1341** represents the attachment depth of the leaflet capture subassembly **1335** to the native leaflet. In some embodiments, the tension of the artificial chordae **1344** may be adjusted at the tuning mechanism **1342**. In some embodiments, the artificial chordae **1344** may be attached to a

muscle using an anchor subassembly **1323**.

(44) FIG. **14** depicts an illustrative pulley-based attachment and tuning mechanism in accordance with an embodiment. As shown in FIG. **14**, the artificial chordae device **1422** may include a pulley tuning mechanism. In some embodiments, the tuning mechanism may include a pulley **1442** that may be controlled by a torque cable **1446** resident in a delivery catheter. In some embodiments, the torque cable **1446** may be attached to the pulley **1442** via a connection feature **1447**. In some embodiments, the leaflet capture assembly **1435** may include a stress relief pad **1438** that enables forces resulting from attachment to the native leaflet **1404** to be distributed. In such embodiments, distributing the attachment forces may prevent damage to the native leaflet **1404** and encapsulate the assembly **1435** after healing. In some embodiments, the artificial chordae **1444** may be attached to the muscle using a barbed anchor subassembly **1423**. The artificial chordae **1444** may be tuned so that the artificial chordae replace the operation of the damaged chordae **1406**.

(45) FIG. **15** depicts an illustrative knot-based attachment and tuning mechanism in accordance with an embodiment. As shown in FIG. **15**, an artificial chordae device **1522** may include a tuning mechanism that includes a knot **1542**. In some embodiments, the knot **1542** may be adjustable and may be used to tune the artificial chordae device **1522** to the correct length. In some embodiments, the knot **1542** may be tuned to the correct length through the delivery catheter **1545**. In some embodiments, the knot **1542** may be locked to a final length through movement of a proximal section of a tightening suture **1543**. In some embodiments, the artificial chordae device **1522** may be attached to the posterior leaflet **1504** and anchored via a screw-based anchor **1523** to the papillary muscles **1503** to repair the damaged chordae **1506**.

(46) FIG. **16** depicts an illustrative knot-based attachment and tuning mechanism in accordance with an embodiment. As shown in FIG. **16**, a plurality of artificial chordae **1644** are implanted in the left ventricle **1602** to properly repair the damaged chordae **1606**. In some embodiments, each artificial chordae may be attached to the posterior leaflet **1604** via a suture **1643** and anchored to the myocardium via a screw-based anchor **1623**. In some embodiments, the length of each of the artificial chordae **1644** may be adjusted independently by adjusting, for example, the corresponding knot-based tuning mechanism. In some embodiments, adjustments to the tuning mechanism may be performed from the delivery catheter.

(47) In some embodiments, multiple devices may be attached to the posterior leaflet or the anterior leaflet. In such embodiments, an operator may control the distance between adjacent attachments to leaflet mechanisms. In another embodiment, one or more pads may be used for stress relief of the attachment. In some embodiments, the pads may be large patches that could support a plurality of attachment mechanisms **1638**. In alternate embodiments, smaller patches may be used that only support a single mechanism **1638**.

(48) FIG. **17** depicts another illustrative anchor-based attachment and tuning mechanism in accordance with an embodiment. As shown in FIG. **17**, artificial chordae **1744** may be implanted to replace damaged chordae **1706** in the left ventricle **1702**. In some embodiments, the artificial chordae **1744** may be attached to the posterior leaflet **1704** in pairs on stress relief pads **1738**. In some embodiments, the artificial chordae **1744** may be anchored to the myocardium below the papillary muscles **1703**. In some embodiments, the tuning mechanism may include one or more torque cables **1746** controlled via the delivery catheter **1745**. In some embodiments, a barbed assembly **1723** may be used to anchor the artificial chordae **1744**.

(49) The length of each of the artificial chordae may be from about 20 mm to about 120 mm. For example, the length of an artificial chordae may be tuned to 20 mm, 30 mm, 40 mm, 50 mm, 60 mm, 70 mm, 80 mm, 90 mm, 100 mm, 110 mm, 120 mm, or to a length within a range between any two of these endpoints. The variation in the length of the artificial chordae may be used to address variations in ventricle size in a patient population as well as to allow the operator to make a decision as to where to anchor the assembly. For example, different operators may choose to anchor the assembly near to the apex, while other operators may choose to anchor the assembly near to the

papillary muscle. Such decisions may be based on the anatomy of the patient and/or operator preference.

(50) In some embodiments, the design of a chordae replacement device may allow loading the device into a low profile shaft having an outer diameter that is, for example, less than or equal to 13 mm. In some embodiments, the chordae replacement device and the associated delivery system may enable delivery of a plurality of devices, such as the leaflet capture assembly, anchor assembly, and artificial chordae. In some embodiments, the plurality of devices may be delivered consecutively or simultaneously. In some embodiments, the plurality of devices may be housed in parallel inside the catheter prior to delivery. In some embodiments, the plurality of devices may be housed consecutively within the catheter prior to delivery. In some embodiments, only one of the plurality of devices may be loaded into the catheter at a time. In some embodiments, the control mechanism (e.g., the torque cable **1746**) may be retained within the catheter.

(51) The leaflet capture assembly may be configured to attach to the native leaflets of the tricuspid valve or the mitral valve. When used in the mitral position, the leaflet capture assembly may lean in against the edges of the mitral leaflets, anterior or posterior, according to the area that requires treatment. In some embodiments, the attachment mechanism may enable tuning of the tension on the artificial chordae that attaches leaflets to the myocardium. In some embodiments, the attachment mechanism may enable precise positioning with respect to the leaflet. In some embodiments, the attachment mechanism may include a padded surface to distribute the forces and/or to encourage encapsulation of the assembly into the leaflet.

(52) One of ordinary skill in the art will be aware that any combination of the various components described herein and equivalents may be used for the construction of a particular device based on this disclosure. In other words, the devices depicted herein are merely illustrative of the types of devices that may be constructed according to the teachings of this disclosure and are not meant to be limited to these illustrative embodiments.

(53) FIG. **18** depicts an illustrative anchor subassembly for an artificial chordae in accordance with an embodiment. The anchor assembly **1823** may be configured to attach to the myocardium. In some embodiments, the anchor assembly **1823** may include a self-expanding stent comprising a shape memory material (e.g., Nitinol). In some embodiments, the anchor assembly **1823** may be cut from a tube or sheet and/or use a pattern that allows crimping and expanding. In one embodiment, the anchoring assembly **1823** may be constructed from a wire formed into a coiled geometry configured to be screwed into the tissue. The coiled portion **1824** of the anchor assembly **1823** may be configured to penetrate into tissue and maintain an anchoring force during implantation and in vivo life expectancy.

(54) In some embodiments, the anchor assembly **1823** may feature an interface portion **1825** that is configured to enable an attachment with a catheter during implantation. The interface portion **1825** may include an attachment **1826** to the delivery catheter that allows transmission of torque forces. The torque forces may be applied to the attachment **1826** to rotate the anchor assembly **1823** and enable the coiled portion **1824** to penetrate into tissue. In some embodiments, axial forces may also be applied to advance the anchor assembly **1823** into or retract the anchor assembly from tissue. In some embodiments, the interface portion **1825** may include one or more holes **1827** configured to attach artificial chordae to the anchor assembly **1823**. Illustrative coil-based anchor assemblies are further depicted in FIGS. **10-13, 15, and 16**.

(55) FIG. **19** depicts an illustrative inner member of an anchor subassembly in a deployed configuration in accordance with an embodiment. In some embodiments, an anchor subassembly may include an outer member (depicted in FIG. **21**) and an inner member **1928**. The inner member may include a tube having a specific pattern that allows two different configurations as depicted in FIGS. **19 and 20**. As shown in FIG. **19**, a plurality of anchors **1929** may be positioned on a first end of the anchor subassembly **1928**. The plurality of anchors **1929** may expand radially to anchor into tissue. In some embodiments, the opposite end **1930** of the anchor subassembly **1928** may provide

an interface for attachment to the artificial chordae.

(56) FIG. **20** depicts the illustrative inner member of the anchor subassembly of FIG. **19** in a non-deployed configuration. As shown in FIG. **20**, the anchors **2029** of the anchor subassembly **2028** may be aligned parallel with the interface **2030** of the anchor subassembly. The anchor **2029** may be in the non-deployed configuration when housed within the outer member of the anchor subassembly as discussed further in reference to FIGS. **21** and **22**.

(57) In some embodiments, the inner member of the anchor subassembly may be constructed from one or more metals and/or an alloy. In one embodiment, the inner member is constructed from one or more metals that are biocompatible and are configured to permit the device to transform between the two configurations. For example, the inner member may comprise one or more of cobalt, chrome, stainless steel, or Nitinol.

(58) FIG. **21** depicts an illustrative outer member of an anchor subassembly in accordance with an embodiment. As shown in FIG. **21**, an outer member **2133** of an anchor assembly may comprise a tube with a plurality of windows **2132**. In some embodiments, each of the plurality of windows may be located radially. Different window locations are contemplated within the scope of this disclosure. In some embodiments, the outer member **2133** may include a distal portion **2134** having a sharp edge. The sharp edge of the distal portion **2134** may be configured to enable the outer member to penetrate into tissue. In some embodiments, the outer member **2133** may further include a proximal section **2131**. In some embodiments, the proximal section **2131** of the outer member **2133** may be used for attachment to the delivery system and/or to the artificial chordae.

(59) In some embodiments, the outer member **2133** of the anchor subassembly may be constructed from one or more metals and/or an alloy. In one embodiment, the outer member **2133** is constructed from one or more metals that are biocompatible. For example, the outer member **2133** may comprise one or more of cobalt, chrome, stainless steel, or Nitinol. In some embodiments, the outer member **2133** may comprise a plastic, such as polyetheretherketone, other materials, or other combinations of materials that are biocompatible.

(60) FIGS. **22** and **23** depict an illustrative anchor subassembly in a non-deployed configuration **2223** and a deployed configuration **2323**, respectively, in accordance with an embodiment. In some embodiments, the inner member **2228/2328** may include a plurality of anchors **2229/2329** and the artificial chordae interface **2230/2330**. In some embodiments, the inner member may be inserted into the outer member **2233/2333**. In some embodiments, the outer member **2233/2333** may include a plurality of anchor windows **2232/2332**, a control interface **2231** to the delivery system, and a distal end **2234/2334**.

(61) In some embodiments, transitioning between the non-deployed configuration and the deployed configuration may be controlled via the delivery catheter by advancing, retracting, and/or rotating the catheter. In some embodiments, transitioning between the two configurations may further be performed with the outer member. For example, linear movement of the inner member **2228/2328** with respect to the outer member **2233/2333** may cause the plurality of anchors **2229/2329** to transition between the two configurations. In some embodiments, movement of the inner member **2228/2328** may cause the outer member **2233/2333** to deflect the anchors through the windows **2232/2332** mechanically. In some embodiments, the inner member **2228/2328** may be manufactured from a shape memory material, such as Nitinol, and movement of the inner member with respect to the outer member **2233/2333** may cause the anchors to transition to the deployed configuration. In some embodiments, the control interface **2231** may be designed to allow an easy and safe release after the device is anchored to the tissue.

(62) In some embodiments, the inner member may be manufactured by machining, grinding, and/or laser cutting. In some embodiments, the distal ends **2234/2334** of the anchors **2229/2329** may be sharpened or grinded to enable tissue penetration. In some embodiments, the anchors **2229/2329** may be aligned in a specific direction. In some embodiments, the anchors **2229/2329** may be aligned in two opposing directions. In some embodiments, the anchors **2229/2329** may be aligned

in a plurality of directions.

(63) Any number of anchors **2229/2329** may be used within the scope of this disclosure. In some embodiments, the distal ends **2234** may be inside the corresponding windows **2232** to assure deployment through the windows when the inner member **2228** is within the outer member **2233** in the non-deployed configuration. In some embodiments, the inner member **2228** may be stored within the catheter and advanced into the outer member **2233** separately. In some embodiments, the inner diameter of the outer member **2233** may be in a range of about 0.8 mm to about 4 mm. In some embodiments, the outer diameter of the inner member **2228** may be in a range of about 0.8 mm to about 4 mm, but in any event is less than the inner diameter of the outer member **2233** to allow movement of the inner member therein.

(64) For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (for example, “a” and/or “an” should be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

(65) In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should be interpreted to mean at least the recited number (for example, the bare recitation of “two recitations,” without other modifiers, means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, et cetera” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (for example, “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, et cetera). In those instances where a convention analogous to “at least one of A, B, or C, et cetera” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (for example, “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, et cetera). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

(66) As will be understood by one skilled in the art, for any and all purposes, such as in terms of providing a written description, all ranges disclosed herein also encompass any and all possible subranges and combinations of subranges thereof. Any listed range can be easily recognized as sufficiently describing and enabling the same range being broken down into at least equal halves, thirds, quarters, fifths, tenths, et cetera. As a non-limiting example, each range discussed herein can be readily broken down into a lower third, middle third and upper third, et cetera. As will also be understood by one skilled in the art all language such as “up to,” “at least,” and the like include the number recited and refer to ranges that can be subsequently broken down into subranges as discussed above. Finally, as will be understood by one skilled in the art, a range includes each individual member. Thus, for example, a group having 1-3 cells refers to groups having 1, 2, or 3 cells. Similarly, a group having 1-5 cells refers to groups having 1, 2, 3, 4, or 5 cells, and so forth.

(67) Various of the above-disclosed and other features and functions, or alternatives thereof, may be combined into many other different systems or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations or improvements therein may be subsequently

made by those skilled in the art, each of which is also intended to be encompassed by the disclosed embodiments.

Claims

1. A method of repairing a native chordae of a patient using an artificial chordae, the method comprising: inserting the artificial chordae into the patient using a delivery system, wherein the artificial chordae is inserted in a non-deployed configuration; delivering the artificial chordae to a desired position within the patient; causing the artificial chordae to transition from the non-deployed configuration to a deployed configuration; anchoring the artificial chordae to a myocardium of the patient via an anchoring assembly comprising an inner member comprising a barbed anchor and an outer member comprising a plurality of windows, the inner member positioned within the outer member; attaching the artificial chordae to a leaflet of the native chordae at an attachment location via a leaflet capture subassembly coupled to the artificial chordae, the leaflet capture subassembly comprising an upper jaw and a lower jaw configured lock the leaflet between the upper jaw and the lower jaw; and tuning the artificial chordae to a desired tension, wherein the artificial chordae is anchored to the myocardium by deploying the barbed anchor of the inner member through the plurality of windows of the outer member and into the myocardium to secure the artificial chordae to the myocardium.
 2. The method of claim 1, further comprising delivering one or more additional devices to the desired position.
 3. The method of claim 2, wherein the one or more additional devices comprise the anchoring assembly and the leaflet capture subassembly.
 4. The method of claim 2, wherein the artificial chordae and the one or more additional devices are delivered via a single catheter.
 5. The method of claim 2, wherein the artificial chordae and the one or more additional devices are delivered in parallel within a single catheter.
 6. The method of claim 2, wherein the artificial chordae and the one or more additional devices are delivered in series within a single catheter.
 7. The method of claim 2, further comprising tuning the one or more additional devices and the artificial chordae simultaneously.
 8. The method of claim 1, further comprising installing a stress relief pad at the attachment location on the leaflet of the native chordae, wherein the stress relief pad is configured to lower stress on the leaflet and assist in healing thereof.
 9. The method of claim 8, further comprising delivering one or more additional devices to the desired position, wherein the artificial chordae and the one or more additional devices are configured to share use of the stress relief pad.
 10. The method of claim 1, wherein the artificial chordae is tuned via a tuning mechanism, the tuning mechanism comprising a pulley interfaced to a torque cable.
 11. The method of claim 1, wherein the artificial chordae is tuned via a tuning mechanism, the tuning mechanism comprising a knot in a tightening suture that is configured to be adjusted to a desired length and locked to the desired length through movement of a proximal section of the tightening suture.
 12. The method of claim 1, wherein the barbed anchor is configured to be delivered to an anchoring location in a non-deployed configuration and then deployed radially into a tissue of the myocardium to secure the artificial chordae thereto.
 13. The method of claim 12, wherein the outer member is configured to: lock the barbed anchor in place during delivery; and control deployment of the barbed anchor through movement of the delivery system.
-

